

Abstract Algebra - Stockholm University 2025

1. Groups

1.1. Conjugation, centralizer, center, normalizer

For $g \in G$, the conjugation map $c_g(x) = gxg^{-1}$ is an automorphism of G (an *inner* automorphism). For a subset $S \subseteq G$, the centralizer is

$$C_G(S) = \{g \in G : gs = sg \text{ for all } s \in S\}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{g \in G : gx = xg \ \forall x \in G\}$$

(always a characteristic, hence normal, subgroup). For a subgroup $H \leq G$, the normalizer is

$$N_G(H) = \{g \in G : gHg^{-1} = H\},$$

the largest subgroup of G in which H is normal (so $H \trianglelefteq N_G(H)$). One always has $C_G(H) \leq N_G(H)$. The number of conjugates of H in G equals $[G : N_G(H)]$.

1.2. Normal subgroups and quotients

A subgroup $N \leq G$ is normal (write $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$. Equivalently, every left coset equals the corresponding right coset: $gN = Ng$. If $N \trianglelefteq G$, the set of cosets $G/N = \{gN : g \in G\}$ is a group with $(gN)(hN) = (gh)N$.

1.3. Cosets and index

For $H \leq G$ and $g \in G$, the left (resp. right) coset is $gH = \{gh : h \in H\}$ (resp. $Hg = \{hg : h \in H\}$). Left cosets are either disjoint or equal and they partition G . Each left coset gH has $|H|$ elements via the bijection $H \rightarrow gH, h \mapsto gh$. The *index* of H in G is the number of left cosets, denoted $[G:H]$.

1.4. Lagrange's Theorem

If G is finite and $H \leq G$, then

$$|G| = [G:H] \cdot |H|.$$

Proof sketch. G is the disjoint union of $[G:H]$ left cosets, each of cardinality $|H|$.

Consequences

- $|H| \mid |G|$ for every $H \leq G$.
- For $g \in G$, the order $|g|$ divides $|G|$; hence $g^{|G|} = e$.
- Index multiplicativity: for $K \leq H \leq G$ (with G finite), $[G:K] = [G:H][H:K]$.
- If $[G:H] = 2$, then $H \trianglelefteq G$ (there are only two cosets, so gHg^{-1} must be H).

Groups of prime order are cyclic. If $|G| = p$ (prime) and $g \in G$, $g \neq e$, then $|\langle g \rangle| = |g| \mid p$ by Lagrange, so $|g| = p$ and $\langle g \rangle = G$. Hence G is cyclic and $G \cong \mathbb{Z}/p\mathbb{Z}$.

1.5. Isomorphism theorems

- (1) **First isomorphism theorem.** For $\varphi : G \rightarrow H$ a homomorphism,

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

- (2) **Second isomorphism theorem.** For $A \leq G$ and $N \trianglelefteq G$,

$$A \cap N \trianglelefteq A, \quad AN \leq G, \quad AN/N \cong A/(A \cap N).$$

- (3) **Third isomorphism theorem.** If $K \leq H \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \cong G/H.$$

- (4) **Correspondence (lattice) theorem.** For $N \trianglelefteq G$, the map

$$\{\text{subgroups } H \mid N \leq H \leq G\} \longleftrightarrow \{\text{subgroups of } G/N\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups $H \trianglelefteq G$ with $N \leq H$ to normal subgroups of G/N .

2. Group actions

3. Sylows Theorem

4. Rings

5. Integral -, Unique Factorization - & Principle Ideal Domains

6. Field Extensions